

Seminar Academic Report

Emotions and Maps: The Affective Dimension of Visualisation

ISSonVis 2026 — International Spring School on Visualization

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1. Executive Summary

This report reflects my personal ideas, key takeaways and the sessions, ideas and discussions that I resonated most with at ISSonVis 2026. The seminar offered a perspective on cartography that I genuinely did not know. The seminar went far beyond the technical dimensions of mapmaking and into something much more human or let's call it emotions. Over two intensive days, sessions touched on spatial cognition, human sensing technologies, AI and Generative AI in cartography, and affective design, all while relating it to the core theme of the seminar, emotions and maps. What stayed with me most was that cartography is not just a technical practice, it is a deeply for human and by human. This report captures the sessions and ideas that I enjoyed the most, and reflects on how they have already begun to shift the way I think about geo-visualization within my studies in the Copernicus Master in Digital Earth and beyond.

2. Technical Sessions Takeaways

2.1 Prompt Cartography and LLM-Driven Workflows

Out of everything at the seminar, the session by Ian Muehlenhaus was the one I left thinking about the most. The project itself was simple in concept but genuinely exciting- he had been creating a new web map every single day using generative AI tools and documenting the process, with the goal of turning it into a book. What drew me to it was that it felt less like a research project and more like something born out of real curiosity and love for cartography, while still being completely grounded in where technology is heading. This is something I always wanted to do and inspired me a lot.

The idea at the heart of it was prompt cartography. A new exciting idea of using large language model (LLM) workflows to turn plain-language instructions into complex spatial visualizations, so that creating a web map becomes something anyone who can write a sentence can do. But what I found most thoughtful about it was the argument that this does not make the geo-informatics professional redundant. It shifts the role toward design governance, making sure that what these automated pipelines produce is accurate, ethical, and not quietly persuasive in ways the user never intended. Likewise, the idea of AI making a map isn't something we

need to be afraid but rather to embrace it and use it as a tool. Because the creativity in map-making will always stay yours, despite the technicalities involved.

2.2 Context-Aware Mapping and GeoAI

This was probably the most technically dense theme of the seminar, but also one of the most fascinating. The sessions explored what becomes possible when physiological human sensing, eye-tracking, EEG, electro dermal activity is brought together with geospatial and Earth Observation data. The result is a new kind of map: not a static deliverable but a dynamic interface that can adapt in real time to how a person is cognitively and emotionally experiencing a space. This ultimately results in high usability in maps.

For me, coming from a web GIS background, this reframed what user-centred design can actually mean. It is no longer just about making an interface easy to use, it is about building something that understands and responds to the person using it. That is a significant shift, and one I think will become increasingly relevant in the years ahead.

2.3 Participatory Emotional GIS

The session by Jiří Panek on emotional mapping was another one that interested me the most. He showed multiple maps drawn by hand, with crayons and colour pencils, yet still spatially meaningful. The philosophy behind the maps was to create a platform was free from every technical barrier so that people can focus entirely on the mapping process and emotions, not on how to operate the tool. No downloads, no installation, just mapping supporting human emotions.

What struck me was the idea to prioritize emotional expression over spatial precision. These maps were not about where things are rather about how places feel. That unique reframing of what a map is for opened up something new in me about how I think about the purpose of geo-visualization altogether.

3. Professional Application

The frameworks introduced at ISSonVis 2026 connect directly to my practice and aspirations as a Cartographer and Earth Observation analyst. My core interest lies in cartography, and the seminar sharpened my thinking about human perspective into mapmaking that I had not previously foregrounded: the affective and cognitive experience of the end user.

Specifically, I intend to carry the following insights into my ongoing and future projects:

- Cartography is more human than a technical methodology: Across every session, the discipline kept revealing itself as something deeply tied to perception, emotion, and lived experience, not just technical process.
- Generative AI is a tool to implement creativity: Prompt cartography showed that lowering the barrier to mapmaking also raises new questions about accuracy, ethics, and design intent.
- User-eye testing data opens a new dimension in mapping: Integrating eye-tracking with geospatial data points toward maps that are genuinely responsive to the person using them.
- Simplicity is a design philosophy, not a limitation: Emotion mapping GIS demonstrated that reducing technical complexity can produce richer, more honest spatial narratives.

Taken together, these represent a more holistic approach to spatial engineering, one that holds technical rigor and human experience as equally important design criteria.

4. Conclusion

ISSonVis 2026 brought together professionals from across various academic profiles including psychology, sociology, graphics and cartography that brought together unique perspectives together under the same theme. This enabled me to think cartography not as a technical workflow but an intersection of more human processes working together to make sense of data. It offered a genuinely expansive view of where cartography is heading and of the responsibilities that come with building the tools that will shape how people understand and navigate their world. The speakers brought together various ideas and concepts on mapping all while staying relevant to the core theme of the summer school which is “emotional mapping” was very impressive. Completing the coursework, session engagement, and this academic report fulfils the requirements for the 3 ECTS credits associated with the event, and marks a meaningful point of reflection in my trajectory within the Copernicus Master in Digital Earth program.

